

SESSION 17

Matrices & Linear Transformations

Matrix Operations · Linear Transformations · Determinants · Matrix Inversion

Matrix Definition & Types

Matrix Multiplication (Algebraic View)

Matrix Multiplication (Geometric View)

Determinant & Volume Scaling

Matrix Inverse & Identity

Transpose & Symmetric Matrices

Orthogonal Matrices

Python Matrix Operations

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)



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01 • Matrix Definition & Types

A **matrix** is a 2D grid of numbers. Types include:

- **Diagonal Matrix:** Non-zero elements only along the main diagonal.
- **Identity Matrix (\$I\$):** Diagonal matrix with 1s on the diagonal. Acts as the number 1 for matrices.
- **Symmetric Matrix:** Transpose of the matrix equals the matrix itself: $A = A^T$.

02 • Matrix Multiplication (Algebraic View)

Multiplication $C = AB$ is defined when the number of columns in A matches rows in B . Element $C_{\{i,j\}}$ is the dot product of Row i of A and Column j of B .

MATRIX MULTIPLICATION ELEMENT

$$C_{(i,j)} = \sum_k A_{(i,k)} \cdot B_{(k,j)}$$

03 · Matrix Multiplication (Geometric View)

Geometrically, multiplying a vector by a matrix ($A\mathbf{x}$) represents a **linear transformation** of the vector. The columns of A define where the standard basis vectors land after transformation.

04 • Determinant & Volume Scaling

The **determinant** of a matrix, $\det(A)$, measures how much area (in 2D) or volume (in 3D) scales after linear transformation. If $\det(A) = 0$, the matrix squishes space into a lower dimension.

05 • Matrix Inverse & Identity

The inverse of matrix A reverses the linear transformation:

MATRIX INVERSE PROPERTY

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

An inverse exists only if $\det(A) \neq 0$ (the matrix is non-singular).

06 • Transpose & Symmetric Matrices

The transpose A^T is formed by swapping rows with columns: $(A^T)_{i,j} = A_{j,i}$.

07 • Orthogonal Matrices

An orthogonal matrix preserves lengths and angles, representing pure rotations or reflections:

ORTHOGONAL MATRIX PROPERTY

$$Q^T \cdot Q = Q \cdot Q^T = I \rightarrow Q^T = Q^{-1}$$

08 · Python Matrix Operations

```
import numpy as np

A = np.array([[1, 2], [3, 4]])
B = np.array([[5, 6], [7, 8]])

# Matrix Multiplication
C = np.dot(A, B) # or A @ B
det_A = np.linalg.det(A)
inv_A = np.linalg.inv(A)

print(f"det(A): {det_A:.4f}")
```

09 · Quick Reference Card



Matrices Cheat Sheet

- Matrix multiplication is non-commutative: $AB \neq BA$ in general.
- Transpose product rule: $(AB)^T = B^T A^T$.
- Inverse product rule: $(AB)^{-1} = B^{-1} A^{-1}$.
- Singular Matrix: $\det(A) = 0$ (no inverse exists).